

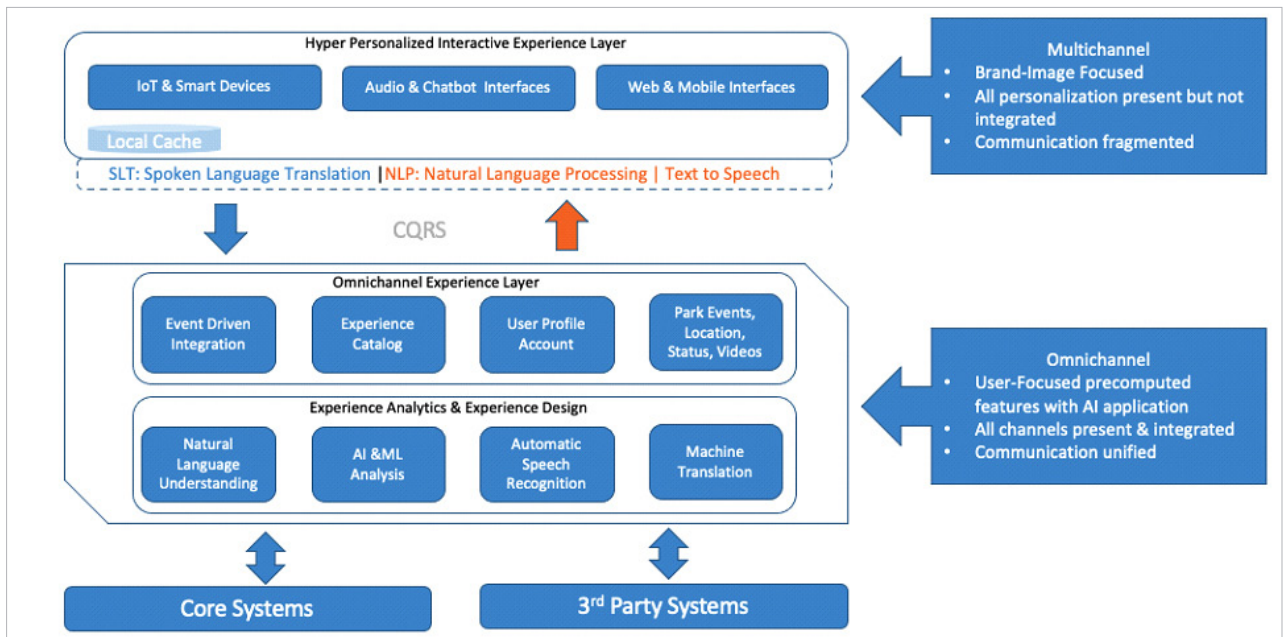


Figure 1: Low latency reference architecture for interactive experience

to ensure that typical outer edge communication issues with centralised and distributed systems are managed. Evolution of 5G has made this journey a little smoother.

A low latency application for an interactive experience

Now that we are through with the fundamentals, let's look at how a low latency application is built in an example use case. The application we are going to build is an interactive experience where the human and machine interaction is expected to occur in real-time or, at least, the human interacting with the machine must not feel a lag in response. One could argue that we already have this in Alexa, Google Assistant, etc. Yes, we do for end consumers for specific search-based responses. Let's look at a reference architecture for an interactive experience application that an organisation tailors for its own end users.

An interactive experience application requires near-real-time responses. This can be done by applying a multichannel framework at the experience layer to build the organisation's brand image.

End user personalisation in a fragmented communication allows for quick consumption and processing of data, limiting any latency aspects in the last step (commonly known as last mile delivery of content). Machine learning models are built with ONNX Runtime, that's built on the Open Neural Network Exchange (ONNX) open standard which has a JavaScript library. Later,

the data required is serialised with open source tools like MLeap or equivalents of it, which is deserialised back for MLeap runtime to power real-time API services in the omnichannel experience layer and beyond.

The middle services layer is designed with an omnichannel framework, where end user focused data is precomputed with the support of

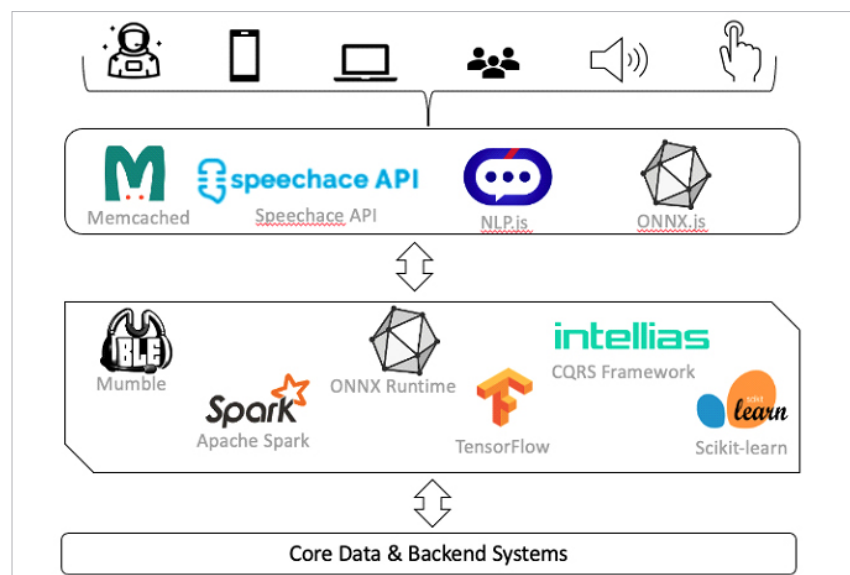


Figure 2: Low latency technical architecture for interactive experience